**Assessment Brief**

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| Module Title: | Applied Statistics & Machine Learning |
| Module Code: | B9BA102 |
| Assessment Title: | Continuous Assessment One |
| Assessment Number: | 1 |
| Assessment Type: | Practical Assignment – Supervised Machine Learning |
| Individual/Group: | Individual |
| Assessment Weighting: | 30% |
| Issue Date: | 28/05/2023 |
| Due Date/Time: | **Deadline – 11.55 pm on Sunday 18/06/2023** |
| Mode of Submission: | MOODLE |

**Learning Outcomes to be assessed**

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| 1. | Analyze a dataset from a problem domain in depth, and select appropriate statistical models, tools, and techniques to derive insights regarding the dataset and domain. |
| 2. | Effectively extract, transform, interrogate, and analyze large datasets. |
| 3. | Construct, refine, interpret, and critically evaluate predictive analytical and machine learning models. |
| 4. | Critically evaluate and utilize hyperparameter search strategies for optimizing machine learning models. |

**Supervised Machine Learning – Classification** **(100 Marks)**

**Dataset**

Each row in ***CustomerChurn.csv*** corresponds to a bank's credit card customer.

Relevant information about this dataset is given below:

**Number of Instances**: 6237
**Number of Attributes**: 15 independent variables + 1 target variable
**Independent Variables**:
*Customer_Age* – Customer age in years
*Gender* – M = Male, F = Female
*Dependent_count* – Number of dependents
*Education_Level* – Highest education level of the customer
*Marital_Status* – Married, Single, Divorced
*Income_Category* – Annual income category of the customer
*Card_Category* – Type of card. Platinum cards offer more reward points than gold cards, which offers more reward points than silver cards. Blue cards offer the least number of reward points on purchases.
*Months_on_book* – Period of relationship (in months) with bank.
*Total_Relationship_Count* – Total number of products held by the customer (e.g., Saving account, Car loan, Credit Card, etc.)
*Months_Inactive* - Number of months inactive in the last 12 months
*Contacts_Count* – Number of customer service contacts in the last 12 months
*Credit_Limit* – credit limit on the credit card
*Total_Revolving_Bal* – Total revolving balance on the credit card (the portion of credit card spending that goes unpaid at the end of a billing cycle)
*Total_Trans_Amt* – Total transaction amount in the last 12 months
*Total_Trans_Ct* – Total transaction count in the last 12 months

**Target Variable:**
*Attrition_Flag* – Two labels - 'Existing Customer', 'Attrited Customer' (Customer who has churned)

**Task**

The bank wants to use a classification model that can predict customer churn. Construct a suitable classification model for the bank by implementing both random forest and support vector classification algorithms in Python.

In addition to providing the python code file, you are required to provide critical analysis of your approach and results in a pdf report.

Your code and analysis should cover the following points:

1. Data Preparation (What steps would you take to prepare your data? Discuss your approach) [20]

2. Model Hyperparameter Tuning (Which hyperparameters would you tune and why? How would you tune them?) [20]

3. Choice of Evaluation Metric (Which metric would be suitable for model evaluation and why?) [20]

4. Overfitting avoidance mechanism (Which mechanism (feature Selection/ regularization) would you use and why?) [20]

5. Results analysis

   a). Which of the two models (random forest or support vector classifier) would you recommend for deployment in the real-world?

   b). Is any model underfitting? If yes, what could be the possible reasons? [20]

**You must submit the following in a zipped folder:**

1. **Critical Analysis Report (.pdf)**
2. **Python Code (.py)**

Naming convention:

Report should be named as –

*Report_Firstname_Surname.pdf*

Code should be named as –

*Code_Firstname_Surname.py*

Zipped folder should be named as –

*Firstname_Surname.zip*

There is no prescribed word-count for the report. It will be assessed on quality, and not quantity of content.

**Assessment Criteria**

Each part will be graded according to the following criteria:

1. Quality of code (correctness and completeness) [Weightage – 40%]

2. Quality of analysis in report (critical analysis of approach, presentation and interpretation of results, conclusion) [Weightage – 60%]

**General Requirements for Students**

PLEASE READ CAREFULLY

1. It is your responsibility to ensure your file is uploaded correctly.
2. Students are required to retain a copy of each assignment.
3. When an assignment is submitted, it is the student's responsibility to ensure that the file is in the correct format and opens correctly.
4. Students should refer to the assessment regulations in their Course Guide.
5. DBS penalizes students who engage in academic impropriety (i.e. plagiarism, Collusion and / or copying). Please refer to the referencing guidelines on Moodle for information on correct referencing.
6. All relevant provisions of the Assessment Regulations must be complied with.
7. Penalties for late submission of assignments are as follows:

a. 25% penalty for assignments submitted within 5 working days of the deadline.

b. No marks for assignments submitted more than 5 working days after the deadline.

Extensions to assignment submission deadlines will be granted in exceptional circumstances only. The appropriate "Application for Extension" form must be used and supporting documentation (e.g. medical certificate) must be attached. Applications for extensions should be made directly to the Programme Coordinator in advance of the deadline date.